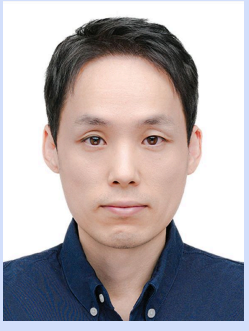

Zhe Hong

Researcher, Environmental Planning Institute, SNU, Korea.

319, Building 82, Seoul National University, Gwanak-gu, Seoul 08826.

Korea (82)010-2910-2015

zhe_hong@126.com

Skills

Statistical analysis tools

Advanced

Stata, R, Python, Mplus

GIS tools

Advanced

Arcgis, Qgis, Geoda

Foreign languages

Korean, English

Fluent

Awards

The Effects of Local Government Capacities on the Improvement of Air Pollution in China

Korean Regional Science Association, Outstanding Conference Paper Award

June 2017

Can machine learning techniques predict energy poverty?

Korea Planning Association Spring Conference, Outstanding Conference Paper Award

April 2021

Summary

Dr.Hong's research interests lie in smart cities, inclusive planning, health city planning, its applications in quantitative analysis, and big data analytics. He obtained a doctoral degree in city planning from Seoul National University, with a dissertation on "Essays on the Relationships Among Urban Environment, Social Capital, and Subjective Well-Being" (in English). To date, he has published a total of 10 papers in representative journals (SSCI and KCI), with works featured in publications such as “Journal of Housing and the Built Environment”, “International Journal of Urban Sciences”, “Social Indicators Research”, “Journal of Korea Planning Association”, and serve as an anonymous reviewer for several SSCI journals, including Social Indicators Research and BMC Public Health.

Education

Seoul National University — **2019.9-2024.2**
Gwanak-gu, Seoul — PhD in City Planning

University of Seoul — **2018.3-2019.8**
Dongdaemun-gu, Seoul — PhD course in Urban Administration

University of Seoul — **2015.9-2017.8**
Dongdaemun-gu, Seoul — Master of Urban Administration

Dankook University — **2013.9-2015.8**
Yongin city, Korea — Bachelor of Marketing

Experience

Environmental Planning Institute, Seoul National University — **2024.4–current**
Bldg. #82, 1 Gwanak-ro, Gwanak-gu, Seoul 08826
Researcher

Brain Korea 21 FOUR Program in Graduate School of Environmental Studies, Seoul National University — **2021.3-2023.8**
Bldg. #82, 1 Gwanak-ro, Gwanak-gu, Seoul 08826
Student Researcher

Seoul National University Asia Center — **2018.9-2020.8**
#101 SNUAC 1Seoul National University, Seoul 08826
Student Researcher

Publications

(SSCI) Hong, Z.*, & Park, I. K. Longitudinal mediating effect of social capital on the relationship between public housing and mental health: evidence from South Korea. — 2023
Journal of Housing and the Built Environment
https://doi.org/10.1007/s10901-023-10015-3

(SSCI) Park, I. K., von Rabenau, B., & Hong, Z.* The effects of ‘sanctuary city’policies on the local economy: empirical evidence from US counties — 2022
International Journal of Urban Sciences
https://doi.org/10.1080/12265934.2022.2093261

(SSCI) Hong, Z.*, & Park, I.K. Is the Well-Being of Neighboring Cities Important to Me? Analysis of the Spatial Effect of Social Capital and Urban Amenities in South Korea. — 2021
Social Indicators Research
https://doi.org/10.1007/s11205-020-02542-1

(SSCI) Hong, Z.*, & Park, I.K. (2018). The effects of regional characteristics and policies on individual pro-environmental behavior in China — 2018
Sustainability
https://doi.org/10.3390/su10103586

(KCI) Zhe Hong*, In Kwon, Park. Comparative Analysis of Energy Poverty Prediction Models Using Machine Learning Algorithms (in English). — 2021
Journal of Korea Planning Association
https://doi.org/10.17208/jkpa.2021.10.56.5.239

(KCI) Zhe Hong*, In Kwon, Park. The Effects of the Revised Housing Allowance System on Housing Cost Burden of Rented Households-An Analysis Using Propensity Score Matching Techniques Combined with a Dynamic Panel Data Model (in Korean). — 2021
Journal of Korea Planning Association
https://doi.org/10.17208/jkpa.2021.04.56.2.22

Projects

Brain Korea 21：Program for Cultivating Innovative Professionals for Sustainable Cities in the Era of Great Transformation — **2020.9-2023.8**
National Research Foundation of Korea, Research participant
https://gsesbk21.snu.ac.kr

Nowadays, cities are facing a huge crisis, and their sustainability is threatened. This research project will continue to pursue the urban transformation in the era of four major crises and opportunities: ① technological and cultural changes, ② environmental and climate change crises, ③ aging and low birth rate crises, ④ social exclusion and inequality crises. Therefore, our vision is to cultivate professionals in urban research related to ① urban smart transformation, ② energy transformation, ③ urban population transformation, and ④ inclusive and shared urban transformation.

A Typology of Korean Cities Based on Urban Inclusivity: Evaluation and Characterization by Type — **2016.3-2017.12**
National Research Foundation of Korea, Research participant

This study aims to classify Korean cities based on the evaluation of their relative inclusivity and to analyze the characteristics of their inclusivity along the typology. To accomplish this, data were collected for 94 cities and counties nationwide as of 2015 according to the Urban Inclusivity Indicators System (UIIS), and a cluster analysis was conducted to classify them into types. In addition, we calculated the comprehensive index score as well as the four dimension scores and the 11 component scores for inclusivity of each type to characterize it. The results show that Korean cities can be classified into three types in terms of inclusivity. Type I consists mainly of mega cities, which has good conditions related to capacity building and empowerment of the socially disadvantaged, and has good interdependence based on formal fair contracts for division of labor. Type II is a type mainly for small and medium sized cities. This type has a good spatial openness in that it has a good inventory of affordable housing such as public rental housing, and a low ratio of housing price to income, while it has medium scores for the indicators along the other dimensions. Type III is a type mainly for rural areas and shows a tendency to be generally opposite to Type I in terms of strengths and weaknesses. The results of this typology and characterization suggest policy areas that each type of city specifically needs to focus on to promote inclusivity.